

This is a copy of a shared ChatGPT chat. It won't be added to memory.

[Report conversation](#)

Give me 12-Month AI Engineer Complete Roadmap (0-Advance), full details structure,

Absolutely. If your goal is to become a **job-ready AI Engineer from zero to advanced in 12 months**, I'd structure the year around **software engineering + ML + deep learning + LLMs + deployment + MLOps**, rather than treating AI as only a collection of courses.

Since you're a CSE student, you can move faster through basic programming and spend more time on ML/AI engineering.

12-Month AI Engineer Roadmap

Zero → Advanced → Job Ready

Target: AI Engineer / ML Engineer / Applied AI Engineer

Duration: 12 months

Recommended effort: 4–6 hours/day, 6 days/week

Total: ~1,200–1,500 focused hours

The Big Picture

MONTH 01

Programming + Git + Linux



MONTH 02

Python for Data + NumPy + Pandas + SQL



MONTH 03

Mathematics + Statistics



MONTH 04

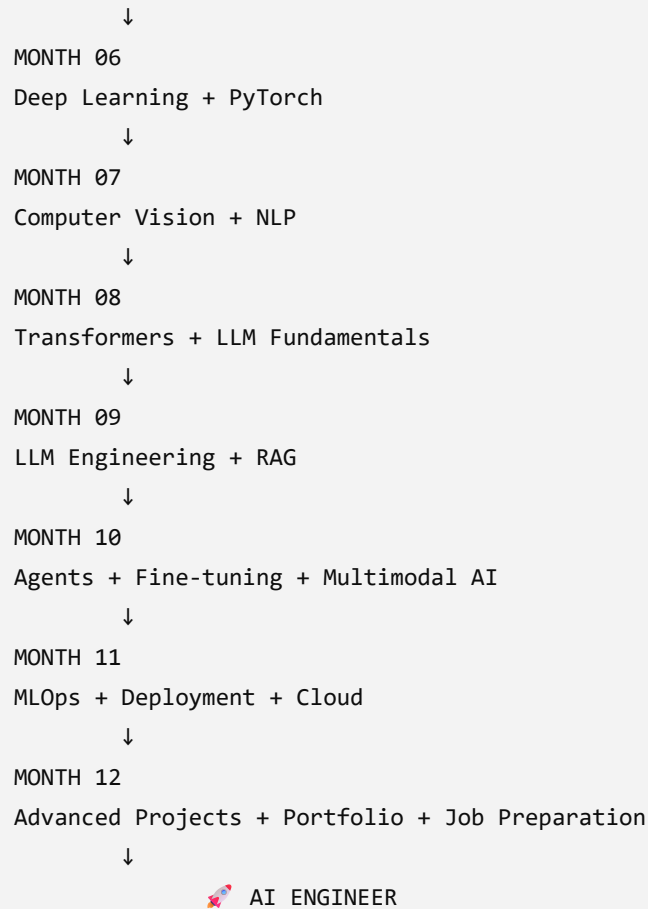
Machine Learning



MONTH 05

Advanced ML + Feature Engineering





MONTH 01 — Programming + Software Engineering Foundations

Goal

Become comfortable writing production-quality Python and working like a software engineer.

Week 1 — Python Fundamentals

Learn:

- Variables
- Data types
- Operators
- Conditions
- Loops
- Functions
- Scope

- Input/output
- Exceptions
- Modules
- Packages

Practice

Build:

1. Calculator
2. Number guessing game
3. Password generator
4. CLI To-Do app
5. File organizer

Week 2 — Intermediate Python

Learn:

Data structures

```
list  
tuple
```

 Python



 Run

Advanced concepts

- List comprehension
- Dictionary comprehension
- Lambda
- `map`
- `filter`
- `zip`
- `enumerate`
- Iterators
- Generators
- Decorators
- Context managers

OOP

Learn:

Class
Object
Constructor
Inheritance
Encapsulation
Polymorphism
Abstraction



Week 3 — Developer Tools

Learn:

Git

```
git init
git add
git commit
git push
git pull
... ..
```

</> Bash

~



Understand:

- GitHub
- Branching
- Pull requests
- `.gitignore`
- README
- Semantic commits

Linux

Learn:

```
ls
cd
pwd
mkdir
rm
cp
mv
cat
grep
find
chmod
```

`</>` **Bash**

```
ssh
```



Week 4 — APIs + Virtual Environments

Learn:

- REST API
- JSON
- HTTP
- GET
- POST
- PUT
- DELETE
- Authentication
- API keys

Python:

```
requests
venv
pip
FastAPI basics
```



Mini Project

AI-ready backend API

Client



FastAPI



Python



JSON response



MONTH 01 PROJECT

Personal Developer API

Build an API containing:

- User system
- Authentication
- CRUD
- Database
- Documentation
- Docker
- GitHub repository

MONTH 02 — Python for AI/Data

Now transition from software engineering → data/AI.

Week 5 — NumPy

Learn:

Array

Shape

Dimension

Indexing

Slicing

Broadcasting

Vectorization

Matrix operations



Important:

```
np.array()  
np.zeros()  
np.ones()  
np.reshape()  
np.transpose()
```

 Python



 Run

Understand why:

Python loops



NumPy vectorization



GPU-friendly computation



Week 6 — Pandas

Learn:

- Series
- DataFrame
- CSV
- Excel
- JSON
- Missing values
- Duplicates
- GroupBy
- Merge
- Join
- Pivot
- Filtering
- Sorting

Practice with:

- Kaggle datasets

- Government datasets
- Open datasets

Week 7 — Data Visualization

Learn:

Matplotlib

Line
Bar
Scatter
Histogram
Boxplot
Heatmap



Seaborn

Learn:

- Distribution plots
- Correlation
- Pairplot
- Categorical plots

Week 8 — SQL

This is extremely important for AI Engineers.

Learn:

```
SELECT  
WHERE  
GROUP BY  
ORDER BY  
JOIN  
HAVING  
CASE
```

</> SQL



Databases:

- PostgreSQL
- SQLite

Understand:

```
AI Engineer
  ↓
Data
  ↓
SQL
  ↓
ML pipeline
```



MONTH 02 PROJECT

End-to-End Data Analysis

Choose a real dataset.

Perform:

```
Raw Data
  ↓
Cleaning
  ↓
EDA
  ↓
Visualization
  ↓
SQL analysis
  ↓
Insights
  ↓
Report
```



Publish on GitHub.

MONTH 03 — Mathematics + Statistics for AI

Don't spend months doing pure mathematics.

Learn **only the mathematics required for AI**.

Week 9 — Linear Algebra

Learn:

Scalars

x



Vectors

$[x_1, x_2, x_3]$



Matrices

$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$



Learn:

- Vector addition
- Dot product
- Matrix multiplication
- Transpose
- Identity matrix
- Inverse
- Rank
- Linear independence
- Eigenvalues
- Eigenvectors

AI connection

Understand:

Neural Networks



Matrix multiplication



Linear algebra



Week 10 — Calculus

Learn:

- Functions
- Derivatives
- Partial derivatives
- Chain rule
- Gradient
- Gradient descent

Most important concept:

Loss



Gradient



Parameter update



Lower Loss



Week 11 — Probability

Learn:

- Probability
- Conditional probability
- Bayes theorem
- Random variables
- Probability distributions
- Expected value
- Variance
- Covariance

Distributions:

Normal
Binomial
Bernoulli
Poisson



Week 12 — Statistics

Learn:

- Mean
- Median
- Mode
- Variance
- Standard deviation
- Correlation
- Covariance
- Sampling
- Confidence intervals
- Hypothesis testing
- p-value
- Central Limit Theorem

MONTH 03 PROJECT



Statistical ML Analysis

Take a real dataset and answer:

What is the distribution?
What variables correlate?
Are groups statistically different?
Are there outliers?
What features matter?



MONTH 04 — Machine Learning Fundamentals

Now the serious AI begins.

Week 13 — ML Fundamentals

Understand:

Dataset
↓
Features
↓
Target
↓
Model
↓
Prediction
↓
Loss
↓
Optimization



Learn:

Supervised Learning

Regression
Classification



Unsupervised Learning

Clustering
Dimensionality Reduction



Week 14 — Regression

Learn:

- Linear Regression
- Polynomial Regression

- Ridge
- Lasso
- ElasticNet

Understand:

$$y = mx + b$$



Then:

Regularization

Bias

Variance

Overfitting

Underfitting



Week 15 — Classification

Learn:

- Logistic Regression
- KNN
- Naive Bayes
- Decision Tree
- Random Forest
- Gradient Boosting
- XGBoost

Week 16 — Evaluation

Classification:

Accuracy

Precision

Recall

F1

ROC-AUC

PR-AUC

Confusion Matrix



Regression:

MAE
MSE
RMSE
 R^2



Learn:

Cross-validation

K-Fold
Stratified K-Fold



MONTH 04 PROJECT



Real-World Prediction System

For example:

Breast cancer classification

Pipeline:

```
Dataset
↓
Cleaning
↓
EDA
↓
Feature Engineering
↓
Train/Test Split
↓
Multiple ML Models
↓
Cross Validation
↓
Hyperparameter Tuning
↓
Evaluation
↓
Explainability
```



MONTH 05 — Advanced Machine Learning

Week 17 — Feature Engineering

Learn:

- Missing-value handling
- Scaling
- Encoding
- Feature selection
- Feature extraction
- Polynomial features
- Outlier treatment

Encoding:

One-hot
Ordinal
Target encoding



Week 18 — Ensemble Learning

Learn deeply:

Bagging
Boosting
Random Forest
AdaBoost
Gradient Boosting
XGBoost
LightGBM
CatBoost



Understand why boosting works.

Week 19 — Unsupervised Learning

Learn:

Clustering

K-Means
DBSCAN
Hierarchical clustering



Dimensionality reduction

PCA
t-SNE
UMAP



Week 20 — Model Explainability

Learn:

Feature importance
SHAP
LIME
Partial dependence



Understand:

Why did the model make this prediction?

MONTH 05 PROJECT

Fraud Detection System

Architecture:

Transaction
↓
Feature Engineering
↓
ML Model



↓
Fraud Probability
↓
REST API
↓
Dashboard

MONTH 06 — Deep Learning

Now learn neural networks properly.

Week 21 — Neural Networks

Understand:

Input
↓
Weights
↓
Bias
↓
Activation
↓
Output



Learn:

- Perceptron
- MLP
- Forward propagation
- Backpropagation
- Loss functions
- Optimizers

Week 22 — PyTorch

Master:

Tensor
Dataset
DataLoader
nn.Module
Optimizer

</> Python



▶ Run

Build models manually.

Week 23 — CNN

Learn:

Convolution
Pooling
Stride
Padding
Channels
Feature maps



Architectures:

LeNet
AlexNet
VGG
ResNet
MobileNet
EfficientNet



Week 24 — Training Deep Networks

Learn:

- Batch normalization
- Dropout
- Weight initialization
- Learning-rate scheduling
- Early stopping

- Data augmentation
- Transfer learning

MONTH 06 PROJECT

Image Classification System

Example:

```
Image
↓
Preprocessing
↓
ResNet / EfficientNet
↓
Prediction
↓
Confidence
↓
FastAPI
↓
Web Interface
```



MONTH 07 — Computer Vision + NLP

This month gives you two important AI domains.

Week 25 — Computer Vision

Learn:

```
Image classification
Object detection
Image segmentation
OCR
Image embeddings
```



Frameworks:

OpenCV
PyTorch
Ultralytics
Transformers



Understand:

YOLO
Faster R-CNN
Mask R-CNN
SAM



Week 26 — Advanced CV

Build:

Object Detection

Image
↓
YOLO
↓
Bounding boxes
↓
Class
↓
Confidence



Learn:

- IoU
- NMS
- mAP

Week 27 — NLP Fundamentals

Learn:

Tokenization
Stopwords
Stemming



Lemmatization
TF-IDF
Bag of Words
Word embeddings

Then:

Word2Vec
GloVe
FastText



Week 28 — NLP Deep Learning

Learn:

RNN
LSTM
GRU
Attention



Understand why Transformers replaced much of the traditional RNN pipeline.

MONTH 07 PROJECT



Multimodal Document Analyzer

Input:

PDF
Image
Text



Output:

OCR
↓
Text extraction
↓
Classification
↓
Entity extraction



↓
Summary

MONTH 08 — Transformers + LLM Fundamentals

This is one of the most important months.

Week 29 — Attention

Understand mathematically:

Query
Key
Value



and:

$\text{Attention}(Q, K, V)$



Understand:

- Self-attention
- Cross-attention
- Multi-head attention

Week 30 — Transformer Architecture

Understand:

Embedding
↓
Positional Encoding
↓
Multi-Head Attention
↓
Feed Forward
↓
LayerNorm





Residual Connection

Learn:

Encoder
Decoder
Encoder-Decoder
Decoder-only



Week 31 — LLM Architecture

Understand:

Tokenizer
↓
Tokens
↓
Embeddings
↓
Transformer
↓
Logits
↓
Probability
↓
Next Token



Learn:

- BPE
- SentencePiece
- Context window
- Temperature
- Top-k
- Top-p
- KV cache

Week 32 — Hugging Face

Master:

Transformers
Datasets
Tokenizers
Accelerate
PEFT



Work with open models.

MONTH 08 PROJECT

Build a Mini LLM

Don't attempt GPT-4 scale.

Implement:

```
graph TD;
  A[Tokenizer] --> B[Dataset];
  B --> C[Transformer];
  C --> D[Training];
  D --> E[Generation];
```



Even a small language model teaches enormous amounts.

MONTH 09 — LLM Engineering + RAG

Now shift from AI research → AI Engineering.

Week 33 — LLM APIs

Learn how to integrate modern LLM APIs.

Architecture:

Application



Prompt



LLM API



Response



Learn:

- System prompts
- User prompts
- Structured output
- Function/tool calling
- Streaming
- Token management

Week 34 — Embeddings

Understand:

Text



Embedding Model



Vector



Learn:

- Semantic similarity
- Cosine similarity
- Vector search

Week 35 — Vector Databases

Learn:

FAISS

Chroma

Qdrant



Weaviate
Pinecone

Understand:

Document
↓
Chunk
↓
Embedding
↓
Vector DB



Week 36 — RAG

Master:

Retrieval-Augmented Generation

Architecture:

Documents
↓
Chunking
↓
Embeddings
↓
Vector DB
↓
User Question → Retrieval
↓
Context
↓
LLM
↓
Answer



Learn:

- Chunking
- Metadata
- Retrieval
- Reranking
- Hybrid search

- Context compression
- Citation
- RAG evaluation

MONTH 09 PROJECT

Production RAG Chatbot

Build a chatbot that can understand:

PDF
DOCX
Web pages
TXT



Features:

- Semantic search
- Citations
- Conversation memory
- Streaming
- Authentication
- File upload
- Retrieval
- Reranking
- Evaluation

This should become one of your **portfolio flagship projects**.

MONTH 10 — Advanced LLM Engineering

Week 37 — Fine-Tuning

Understand:

Pretraining
↓
Instruction tuning



↓
Fine-tuning
↓
Preference optimization

Learn:

LoRA
QLoRA
PEFT



Week 38 — LLM Evaluation

This is often neglected.

Learn:

Accuracy
Groundedness
Faithfulness
Relevance
Toxicity
Hallucination
Latency
Cost



Build evaluation datasets.

Week 39 — AI Agents

Learn:

LLM
↓
Reasoning
↓
Tool
↓
Observation
↓
Action





Result

Tools can include:

Search
Calculator
Database
Python
APIs
File systems



Understand:

- Tool calling
- Agent loops
- Planning
- Memory
- State
- Guardrails

Week 40 — Multimodal AI

Learn systems that handle:

Text
Image
Audio
Video
Documents



Build:

Image → Vision model → Text
Audio → Speech model → Text
Text → LLM → Action



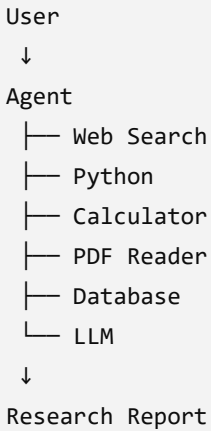
MONTH 10 PROJECT



AI Research Agent

Example:

```
graph TD
    User --> Agent
    Agent --- Tools[Web Search  
Python  
Calculator  
PDF Reader  
Database  
LLM]
    Tools --> Report[Research Report]
```



Include:

- Tool calling
- Citations
- Memory
- Structured output
- Evaluation
- Error handling

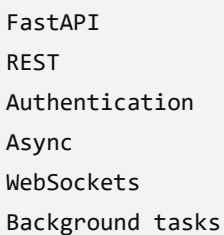
MONTH 11 — MLOps + Deployment

This separates an **AI learner** from an **AI Engineer**.

Week 41 — Backend Engineering

Master:

```
FastAPI
REST
Authentication
Async
WebSockets
Background tasks
```



Learn:

JWT
OAuth basics
Rate limiting
Logging
Error handling



Week 42 — Docker

Master:

```
FROM  
RUN  
COPY  
WORKDIR  
  
</> dockerfile
```



Learn:

Docker
Docker Compose
Container networking
Volumes
Environment variables



Week 43 — CI/CD

Learn:

GitHub Actions
Testing
Linting
Build
Deploy



Pipeline:

Git Push



Tests



Build Docker



Deploy



Monitoring



Week 44 — Cloud

Learn at least one:

AWS

GCP

Azure



Understand:

Compute

Storage

Database

Networking

IAM

Containers

GPU



Week 45 — MLOps

Learn:

Experiment tracking

Model registry

Data versioning

Model versioning

Monitoring

Drift

Retraining



Tools:

MLflow
Weights & Biases
DVC
Docker
Kubernetes basics



MONTH 11 PROJECT

Production ML System

Build:

```
graph TD; Dataset --> TP[Training Pipeline]; TP --> ET[Experiment Tracking]; ET --> MR[Model Registry]; MR --> Docker; Docker --> FastAPI; FastAPI --> Cloud; Cloud --> Monitoring
```



MONTH 12 — Advanced AI Engineering + Career

This is the **capstone** month.

Week 46 — System Design for AI

Learn to design:

AI Chatbot

Frontend



API Gateway



Backend



LLM

├─ Vector DB

├─ Cache

├─ Database

└─ Tools



Understand:

- Scalability
- Latency
- Cost
- Reliability
- Security
- Caching
- Queues
- Rate limiting

Week 47 — AI Security

Learn:

Prompt injection

Jailbreaking

Data leakage

PII

Insecure tool calls

Model abuse

RAG poisoning



Understand:

Guardrails

Input validation



Policy checks



↓
LLM
↓
Output validation

Week 48 — AI Performance Optimization

Learn:

Quantization
Distillation
Pruning
Caching
Batching
Streaming
Speculative decoding



Understand:

Quality
↕
Latency
↕
Cost



Week 49 — Portfolio

You should now have approximately:

Project 1

ML Prediction System

Project 2

Computer Vision System

Project 3

Production RAG

Project 4

AI Agent

Project 5

Production MLOps System

Week 50 — GitHub

Your GitHub should look like:

github.com/yourname



```
|— ml-prediction-system
|— computer-vision
|— rag-platform
|— ai-research-agent
|— mlops-platform
└— mini-transformer
```

Every serious project should have:

README

Architecture diagram

Installation

Usage

Dataset

Model

Evaluation

Screenshots

API documentation

Docker

Demo



Week 51 — Interview Preparation

ML Questions

Know:

- Bias vs variance
- Overfitting
- Regularization

- Gradient descent
 - Cross-validation
 - Feature engineering
 - Ensemble methods
 - Evaluation metrics
-

Deep Learning

Know:

- Backpropagation
 - CNN
 - RNN
 - Attention
 - Transformer
 - BatchNorm
 - Dropout
 - Optimizers
-

LLM

Know:

- Tokenization
 - Embeddings
 - Attention
 - Transformer
 - RAG
 - Fine-tuning
 - LoRA
 - Quantization
 - Agents
 - Tool calling
 - Evaluation
-

Software Engineering

Know:

- API
- Database
- Docker
- Git
- Testing
- CI/CD
- Cloud
- System design

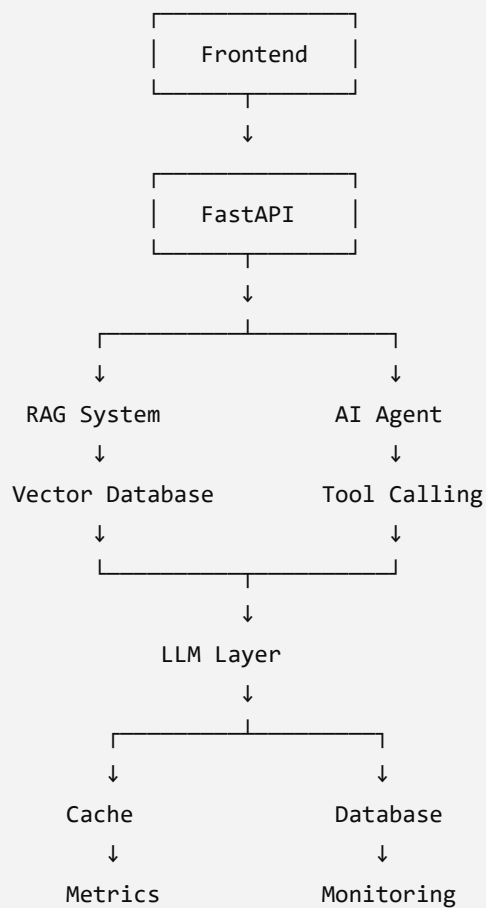
Week 52 — Capstone

Build one large production-grade AI application.

For example:



AI Knowledge Platform





Complete Technology Stack

By the end of 12 months, aim to be comfortable with:

Programming

Python ★★★★★

C++ ★★★

JavaScript/TypeScript ★★★

SQL ★★★★★

Bash ★★★



You don't need to become equally expert in every language.



Data

NumPy

Pandas

Matplotlib

Seaborn

SQL

PostgreSQL



Machine Learning

Scikit-learn

XGBoost

LightGBM

CatBoost



Deep Learning

PyTorch ★★★★★

TorchVision



TensorBoard
Weights & Biases

Computer Vision

OpenCV
YOLO
SAM
Transformers



NLP

NLTK
spaCy
Transformers
Sentence Transformers



LLM

Hugging Face
Transformers
PEFT
LoRA
QLoRA
vLLM
Ollama



RAG

Embeddings
FAISS
Qdrant
Chroma



Hybrid Search
Reranking



Backend

FastAPI
REST API
WebSockets
PostgreSQL
Redis



Deployment

Docker
Docker Compose
GitHub Actions
AWS/GCP/Azure



MLOps

MLflow
W&B
DVC
Monitoring
Model Registry
CI/CD



Architecture

Learn:

Microservices
Event-driven architecture
Message queues
Caching



Load balancing
API gateways
Distributed systems



Recommended Daily Schedule

If you have 5 hours/day:

Hour 1



Theory

ML/DL/LLM concepts



Hour 2



Coding

Implement from scratch

✦ Upgrade



Hour 3



Project

Real-world implementation



Hour 4



Problem solving

ML experiments
Kaggle
research papers



Hour 5



Engineering

Git
Docker
API



deployment
documentation



Weekly Structure

Use:

MON → Theory + Coding
TUE → Theory + Coding
WED → Implementation
THU → Implementation
FRI → Project
SAT → Project + Research
SUN → Review / Rest



The Most Important Learning Rule

Don't do this:

Course
↓
Course
↓
Course
↓
Course
↓
Certificate



Instead:

Learn
↓
Implement
↓
Break
↓
Debug
↓
Build
↓



Deploy



Document



Your Project Progression

Your projects should increase in complexity:

```
LEVEL 1
Python application
    ↓
LEVEL 2
Data analysis
    ↓
LEVEL 3
ML prediction
    ↓
LEVEL 4
Deep learning
    ↓
LEVEL 5
Computer vision
    ↓
LEVEL 6
Transformer
    ↓
LEVEL 7
RAG
    ↓
LEVEL 8
AI Agent
    ↓
LEVEL 9
Production AI
    ↓
LEVEL 10
Scalable AI System
```



What You Should Be Able to Do After 12 Months

You should be able to take:

"We need an AI system that answers questions from 100,000 company documents."

and independently think:

Requirement



Data ingestion



Document processing



Chunking



Embedding



Vector DB



Retrieval



Reranking



LLM



Guardrails



Evaluation



FastAPI



Docker



Cloud



Monitoring



That is the mindset of an **AI Engineer**.



Final Skill Matrix

Skill

Target

Python



SQL

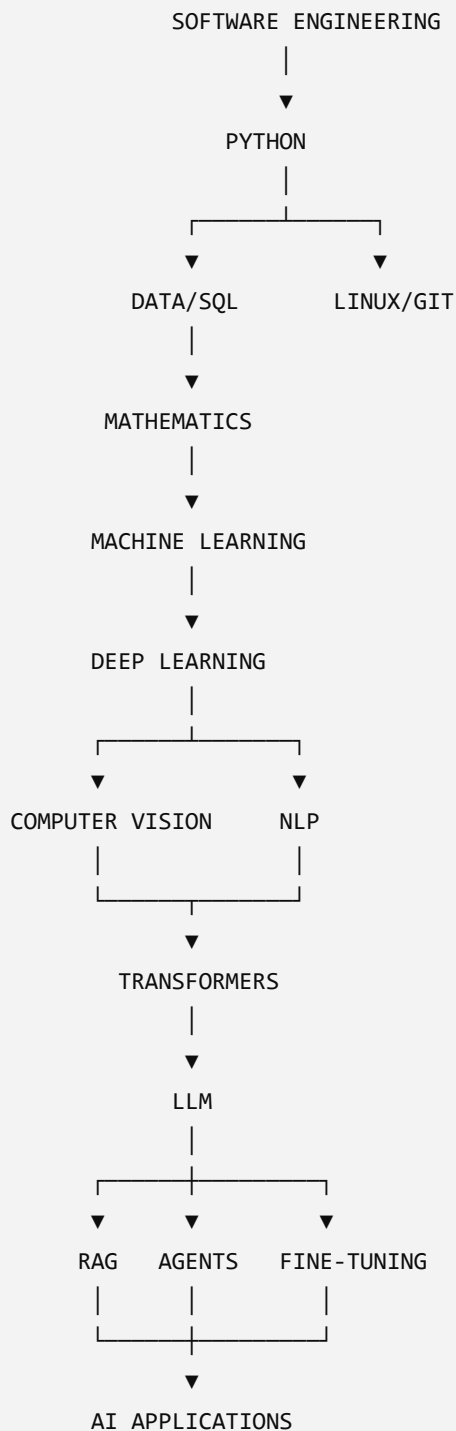


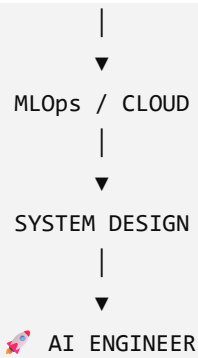
Skill	Target
Git/GitHub	★★★★
Linux	★★★★
Mathematics	★★★★
Statistics	★★★★
Machine Learning	★★★★★
Deep Learning	★★★★★
PyTorch	★★★★★
Computer Vision	★★★★
NLP	★★★★
Transformers	★★★★★
LLM Engineering	★★★★★
RAG	★★★★★
AI Agents	★★★★
Fine-tuning	★★★★
FastAPI	★★★★
Docker	★★★★
Cloud	★★★
MLOps	★★★★
System Design	★★★★
AI Security	★★★



The End Goal

Your learning path should ultimately become:





One important adjustment for your background: because you already have CSE/programming exposure, I would *not* spend the full first two months relearning programming from scratch. You can compress Months 1–2 and use the recovered time for **ML fundamentals, PyTorch, LLM engineering, and larger production projects**. That will give you a much stronger AI-engineering profile than collecting lots of beginner certificates.

If you want, I can:

- ↳ [Expand Month 12: Career and Projects Details](#)
- ↳ [Add Example Code Snippets for Key Technologies](#)
- ↳ [Include Study Plan with Weekly Milestones](#)

ChatGPT can make mistakes. Check important info.

[Cookie preferences](#)